eURC

eURC Point of Sale (React + Node.js) · Hands-on Project 02 · DevOps architecture

Team, workflow and toolchain

Sheet 1 · 2 developers + 1 tester, GitHub flow, reviews, tools

A · Delivery team (config/eurc.env: GITHUB_OWNER, DEV1, DEV2, TESTER)

Developer 1 · lead
DEV1 = GITHUB_OWNER (repo admin)

Owns (CODEOWNERS): app/ web/ infra/ scripts/ · with Tester: .github/

- builds API and React features
- reviews Developer 2's pull requests
- owns Bicep, Azure bootstrap, budget
- backup prod approver
- decides on rollback (S1)

Developer 2
DEV2 (write access)

Owns (CODEOWNERS): app/ web/ infra/ scripts/

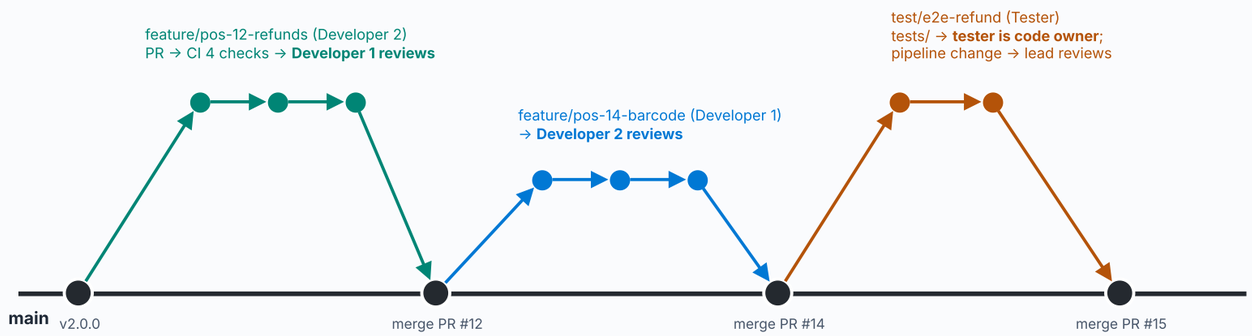
- builds API and React features
- reviews Developer 1's pull requests
- adds unit / component tests with each change
- writes expand-only DB migrations
- cannot merge without a review

Tester (QA engineer)
TESTER (write + prod reviewer)

Owns (CODEOWNERS): tests/ · with lead: .github/

- writes acceptance criteria with the team
- owns smoke, integration, load and Playwright suites
- explores each release on **test**
- **QA sign-off**: approves prod
- files bugs (S1–S4 template)

B · GitHub flow: short-lived branches, pull requests, protected main



Rules on main (ruleset protect-main): pull request only · 1 approval after the last push · code-owner approval · all threads resolved · checks **app, web, infra, scripts** green and branch up to date · no force-push / delete · squash-merge.

C · DevOps lifecycle and toolchain (who does what at each stage)

1 · Plan	2 · Code	3 · Build	4 · Test	5 · Release	6 · Deploy	7 · Operate	8 · Monitor
GitHub Issues feature + bug templates acceptance criteria TEAM.md (RACI, DoD)	VS Code / Codespaces React 19 + Vite 8 Node 22 + Express 5 Bicep · bash	GitHub Actions (CI) npm ci · vite build ESLint · npm audit one zip: API + UI	node:test · Vitest + RTL smoke · integration load gate Playwright E2E	environment prod QA sign-off GitHub release tag change in run summary	OIDC → Azure (no secrets) Bicep what-if + deploy az webapp deploy staging slot → swap	App Service health check rollback workflow delete locks · budget 30-monitor.sh	Application Insights Log Analytics (KQL) 6 alerts + availability DORA metrics
Who: Team	**Who:** Dev 1 · Dev 2	**Who:** automatic	**Who:** Tester + CI	**Who:** Tester (lead backup)	**Who:** automatic	**Who:** Dev 1	**Who:** Team

feedback loop: alerts, bugs, DORA metrics → next sprint's plan

Local inner loop: **npm run dev** in app/ (in-memory API, demo store) + web/ (Vite, proxies /api) — no Azure needed until the pull request.

D · Environments, access and approvals

	local	dev	test	prod
Who uses it	each developer	developers	tester, demos	real stores
Deployed by	npm run dev	CD, every merge	CD, after dev	CD, after QA sign-off
Approval	—	none (main only)	none (main only)	**tester** or lead; not the run's trigger
Data	in-memory demo	test data	test data (E2E cleans up)	live, private endpoints
Azure identity	—	id-eurc-gh-dev-cc	id-eurc-gh-test-cc	id-eurc-gh-prod-cc
UI badge	LOCAL grey	DEV green	TEST amber	PROD blue

E · Definition of Done

- ✓ acceptance criteria automated (unit, component, integration or E2E)
- ✓ CI green; API and React coverage ≥ 80 %
- ✓ reviewed by a code owner, not the author
- ✓ DB change is expand-only (both slots share SQL)
- ✓ no secrets or card data in code, logs or settings
- ✓ deployed to test; Playwright green on desktop + tablet
- ✓ tester signed off; prod gate green after swap

F · Delivery metrics (30-monitor.sh --dora)

Metric	Source
Deployment frequency	successful prod jobs per day
Lead time for changes	commit → live in prod (incl. approval)
Change failure rate	failed prod jobs ÷ attempts
Time to restore	Rollback production runs (≈ 1–2 min)

Project	Drawing	Sheet
eURC POS · P02	EURC-P02-DVO-01	1 of 2
Revision	Standard	Format
1.1 · 2026-10-23	Azure CAF naming	A3 landscape

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CI/CD pipeline — build once, promote dev → test → prod

Sheet 2 · workflows, gates, QA sign-off

① Pull request → CI (.github/workflows/ci.yml · no Azure access)

Developer 1 / 2 or tester feature/* · test/* —push→ Pull request to main

job: app · Node.js API
- ✓ ESLint
- ✓ 31 unit + HTTP tests (node:test)
- ✓ coverage ≥ 80 % lines
- ✓ npm audit (high+)

job: web · React
- ✓ ESLint + react-hooks
- ✓ 17 component tests (Vitest, RTL)
- ✓ coverage ≥ 80 % · vite build
- ✓ npm audit (high+)

job: infra · Bicep
- ✓ bicep build + lint
- ✓ build-params dev / test / prod
- ✓ linter rules as errors

job: scripts · tests
- ✓ ShellCheck
- ✓ actionlint
- ✓ Playwright specs list

Ruleset **protect-main**
4 checks green · 1 approval
code owners · threads resolved
never self-approved
—pass→ **squash-merge to main**

Reviewer by CODEOWNERS: app/ web/ infra/ → the other developer · tests/ → tester · .github/ → lead or tester. A red check blocks the merge.

② Push to main → CD (.github/workflows/cd.yml · OIDC per environment · concurrency: one release at a time)

build

API: lint · tests · coverage
React: lint · tests · **vite build**
validate Bicep (3 envs)
version **2.0.0+<run>.<sha>**
API prod deps + React dist
→ public/ + build.json
eurc-app.zip → artifact

One package (API + UI) for every environment.

—artifact→

dev (automatic)

azure/login — id-eurc-gh-dev-cc
deploy-infra: what-if + Bicep
(only if infra/ changed,
forced, or first deploy)
az webapp deploy (zip)
smoke: 8 checks, waits for
Healthy + expected version
+ React bundle served
reusable: _deploy-env.yml

—pass→

test (automatic)

azure/login — id-eurc-gh-test-cc
deploy · **smoke** 8 checks
integration: store trading day,
13 checks (SQL, Key Vault, Blob)
load: 60 s × 10 workers, sales +
reads; p95 < 1.5 s, err < 1 %
Playwright E2E (tester-owned):
5 journeys × desktop + tablet
JUnit + HTML report → artifact

—pass→

QA sign-off
env prod: tester approves
(lead = backup)
no self-approval

—approve→

prod (after QA sign-off)

azure/login — id-eurc-gh-prod-cc · deploy-infra
record version currently live
1 deploy zip → **staging slot**
2 smoke staging (env=prod, slot=staging)
3 swap staging ↔ production (warm-up /health = 200)
4 verify production (version, slot, health)
5 post-deploy gate: 3 min read-only traffic +
AppRequests failure rate < 2 % (Log Analytics)
6 release annotation in Application Insights
failure after step 3 ⇒ automatic swap back

—pass→

release tag

gh release create
v2.0.0-<run>.<sha>
notes link the run

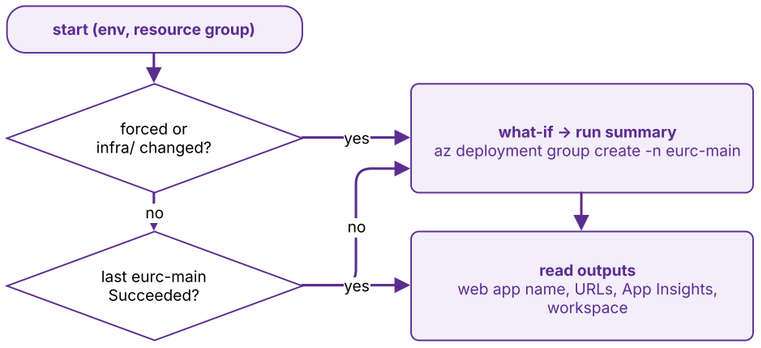
DORA metrics:
30-monitor.sh --dora

× any failed gate stops the promotion; later environments keep the previous version

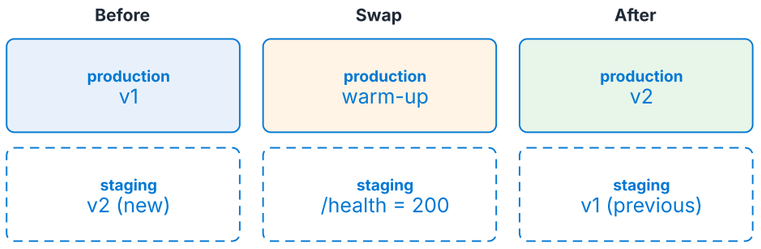
Rollback production (rollback.yml)

manual: reason → approval → swap previous version back → verify (~1–2 min)

③ deploy-infra composite action (.github/actions/deploy-infra)



④ Zero-downtime release with a staging slot



Users stay on production while staging is deployed and tested. The swap only switches routing after the staging instances answer /health. The previous version stays warm in staging, so a rollback is just another swap. EURC_SLOT is a slot setting (sticky); everything else moves with the code. Both slots use the same database ⇒ migrations must be expand-only.

Stage	Gate	Pass condition	On failure
PR	4 checks + review	green · code owner approved	merge blocked
build	API + React tests	48 pass · ≥ 80 % lines each	no artifact
dev	smoke	8/8 · expected version	stop
test	integration · load	13/13 · p95 < 1.5 s	stop
test	Playwright E2E	10/10 (2 devices)	stop
prod	QA sign-off	tester approves	waits
prod	staging smoke	8/8 on staging	no swap
prod	verify + gate	8/8 · failures < 2 %	auto swap back

Secrets in the pipeline: **none**. GitHub holds IDs only; each job gets a short-lived Entra token for one resource group.

Project **eURC POS · P02**	Drawing **EURC-P02-DVO-02**	Sheet **2 of 2**
Revision **1.1 · 2026-10-23**	Standard **Azure CAF naming**	Format **A3 landscape**